

Science First Year topics notes !.

Science Methods

Hypothesis: is an educated guess to problem based on research.

Independent variable: variable that will change.

Dependant variable: variable results depend on that of another.

Controlled variable: variable that will not change but must be constant.

Science Method

Biased experiment is where both sides of an argument are not given equal investigation.

Theory is a hypothesis which is supported with body of evidence.

Cell

Parts of the cell

Cytoplasm=jelly like substance that carries out many cell reactions.

Cell membrane= controls what is going in and out of the cell.

Nucleus=controls the cell activity and structure.

Mitochondria=Energy source for the cell it carries out chemical reaction called respiration.

Cell wall=made from the cellulose permeable involved in support and protection.

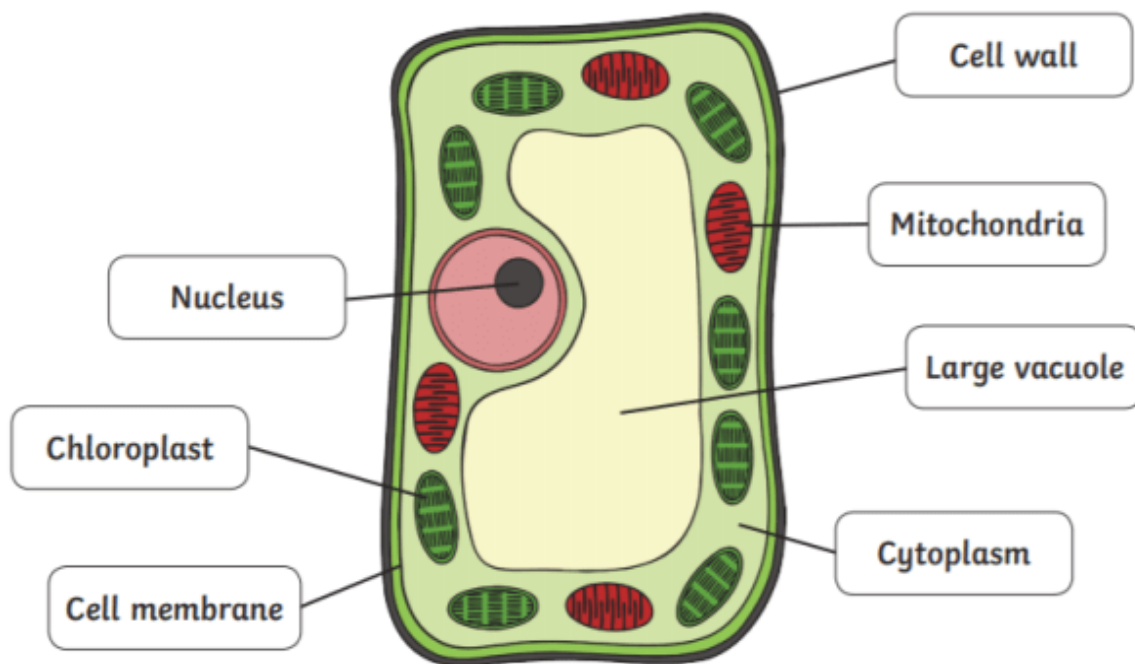
Chloroplasts= containing chlorophyll the site of photosynthesis/allow plant to make food.

Ribosomes=make protein e.g. Enzymes

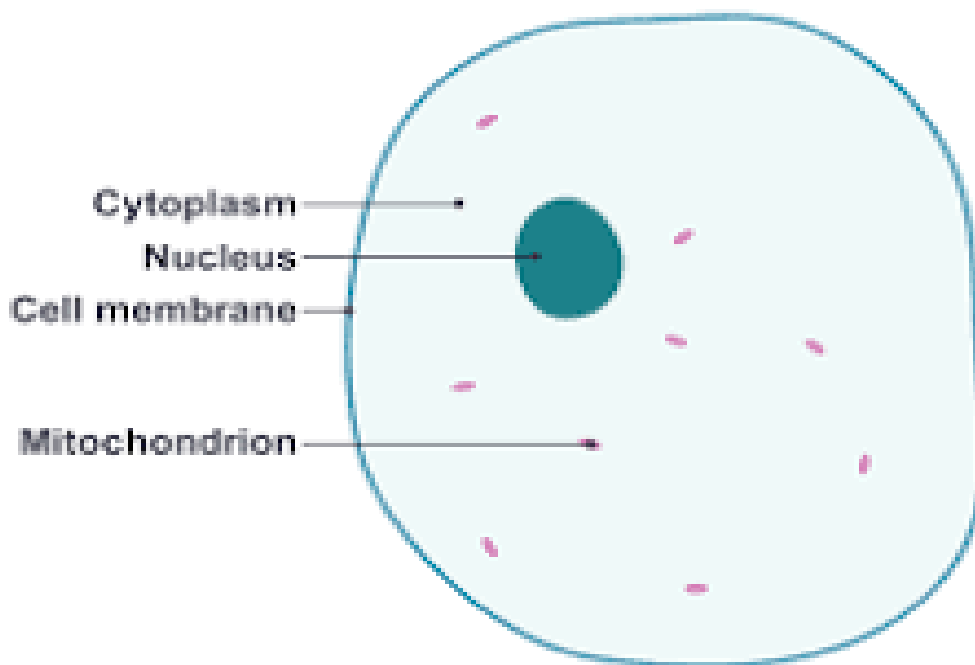
Vacuoles= stores food and waste.

Plant cell

Plant Cell Diagram



Animal cell



Differences between a plant cell and animal cell

Plant

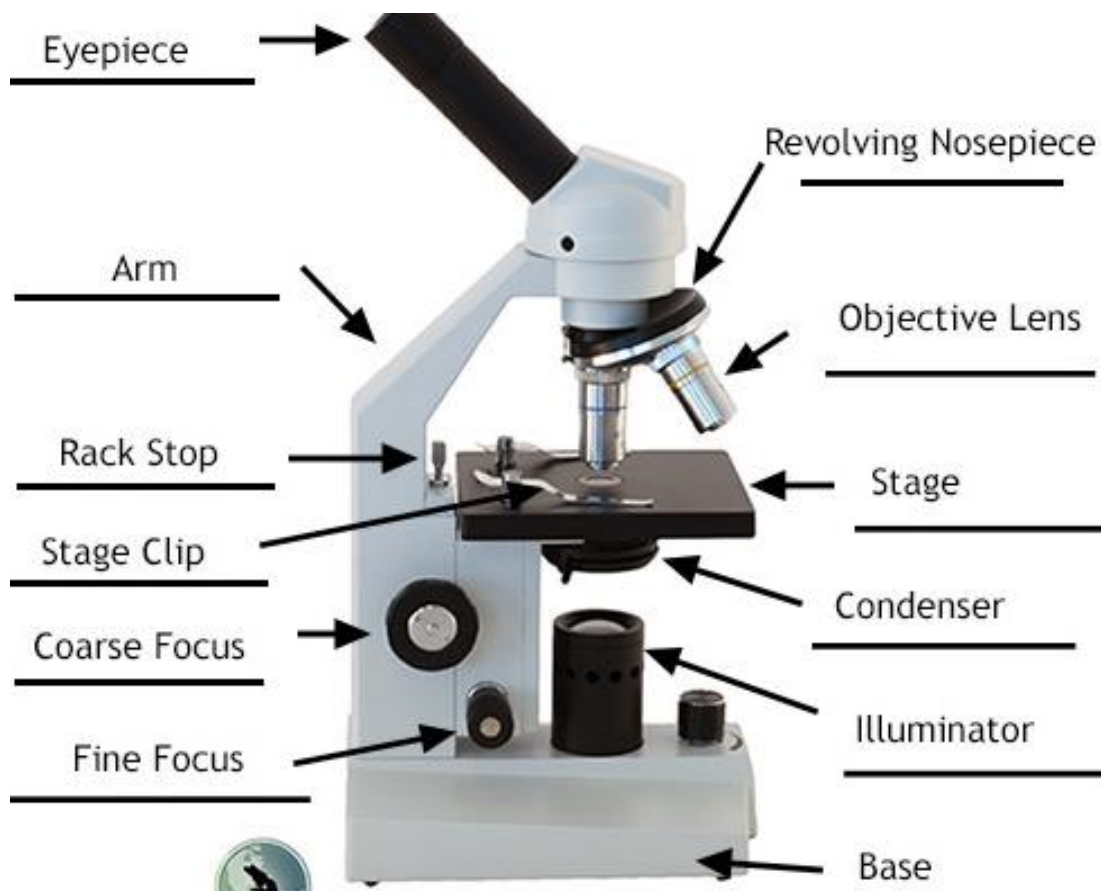
Animal

Chloroplast No Chloroplast

Cell wall No Cell wall

Large vacuoles No Vacuoles

Microscope



7 Parts of the microscope and their functions

Eyepiece lens= magnifies the specimen

Course Focus= to get a rough image

Fine Focus= to get a precise image

Objective lens= magnifies the specimen

Parts of the microscope and their functions

Nosepieces= Allows the object the objectives lens to be changed

Stage clips=Hold the slide in place on the stage

Light source=Projects light though the microscope

Preparing a slide

1. Place a drop of water in the centre of the slide and position the sample on the water, using tweezers. Then add a stain to the sample.
2. At an angle, place one side of the cover slip against the slide, making contact with outer edge of the liquid drop.
3. Lower the cover slowly, avoiding air bubbles.
4. Remove excess water with the paper towel.

Seven characteristics of living things.

Nutrition the way it in a living, things get their food.

Respiration, their release of energy from the food.

Excretion the way in living things get rid of waste.

Growth. The cells divide, which allows all living things grow.

Reproduction the way in living things, produce knew living things.

Movement. Animals move towards food. Plants movement is much slower.

Response the way in living things react to changes in their surroundings.

Investigate preparing the slice of a plant and animal cell.

Method for viewing animal cell and Human cheek cell.

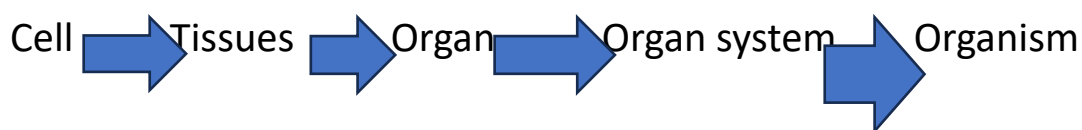
1. Using cotton wool bud, rub the inside of your mouth and smear on the glass slide.
2. Place the drop of methylene blue on top of the smear and allowed to soak it in for 5 minutes.
3. Low over the cover slip slowly using needle from a 45-degree angle to avoid trapping air bubbles.
4. View slide under the light microscope at this low power or 40X focus and then move to the medium to higher to see the cell.

Method for viewing plant cell (onion cell)

Cut the piece of onion and remove a single layer of. Cells and place on the glass slide.

1. Place a few drops of iodine onto the onion layer and allow to soak in.
2. Lower the cover. Slip slowly using the needle at 45° angle to avoid dropping the air bubbles.
3. View slide under the light microscope is 40X focus and then move to medium and higher to see the cell better.

Cell Organisation



Organisation of multicellular organisms

Cell and the smaller working units of the multicellular organisms.

Multicellular organisms are made up of more than one cell

Cell together, form a tissue, for example, skin tissue, a muscle tissue and bone tissue

Tissue together forms again for example stomach and heart.

Organs together form a system for example digestive system and breathing system.

System together form organism for example of fly human animal.

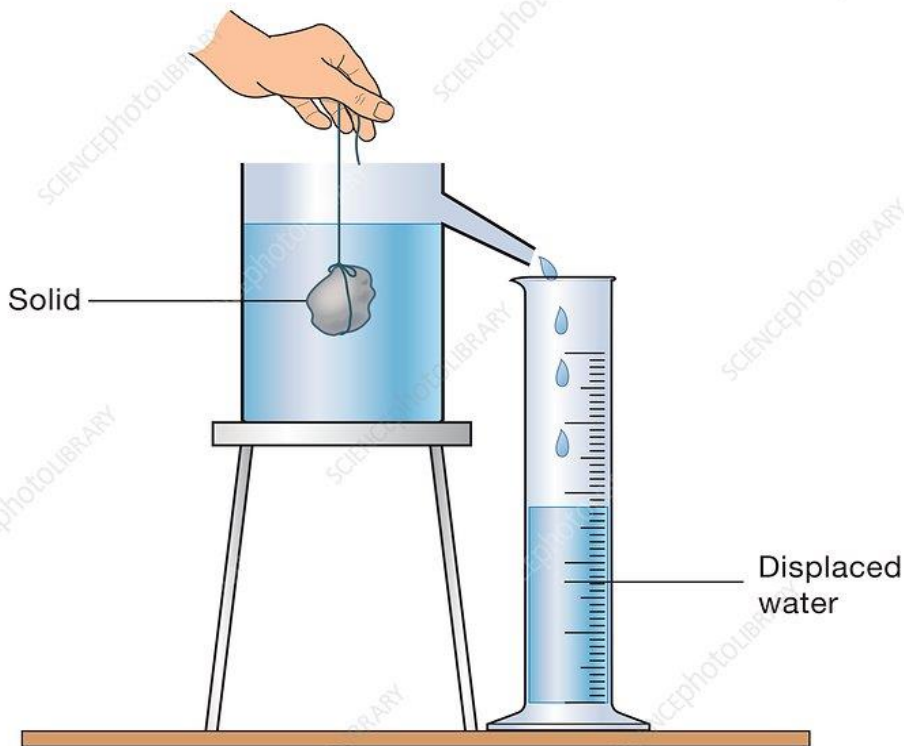
Experiment: Find the volume of the irregular shape

Equipment

- Measuring cylinder
- Stone
- Overflow can/displacement can
- Thread
- Water

Method

1. We place the overflow can in the higher stand than the cylinder, then we put some water in overflow can until every single drop is gone from the nose piece.
2. Tied the rock to the thread and put the cylinder under the vent.
3. During the immersion of rock, part of the water from the overflow can though the vent will be poured into a cylinder.
4. The volume of water that flows out of the overflow can will be equal to the volume of the immersed part of the rock.
5. We look at the measuring cylinder and determine the volume of the water in it. This volume will be equal to the volume of the rock.
6. We must do the experiment 3 times to get the average.



States of matter

There are three states of matter

- Solid
- Liquids
- Gasses

Properties of matter.

Matter is anything. – takes up space and has mass.

Viscosity this describes the flow a rate a liquid.

A special characteristic of gas is compression.

Compression is described if liquids or gasses can be squashed.

Mass: The amount of matter in an object.

Volume: The amount of space an object takes up.

Measuring Matter

1g=1000mg

1kg=1000g

1 tonne=100kg

Mass and Weight

Mass

Amount of matter in a substance

Remains constant

Unit=Kg

Weight

Changes with the force of gravity

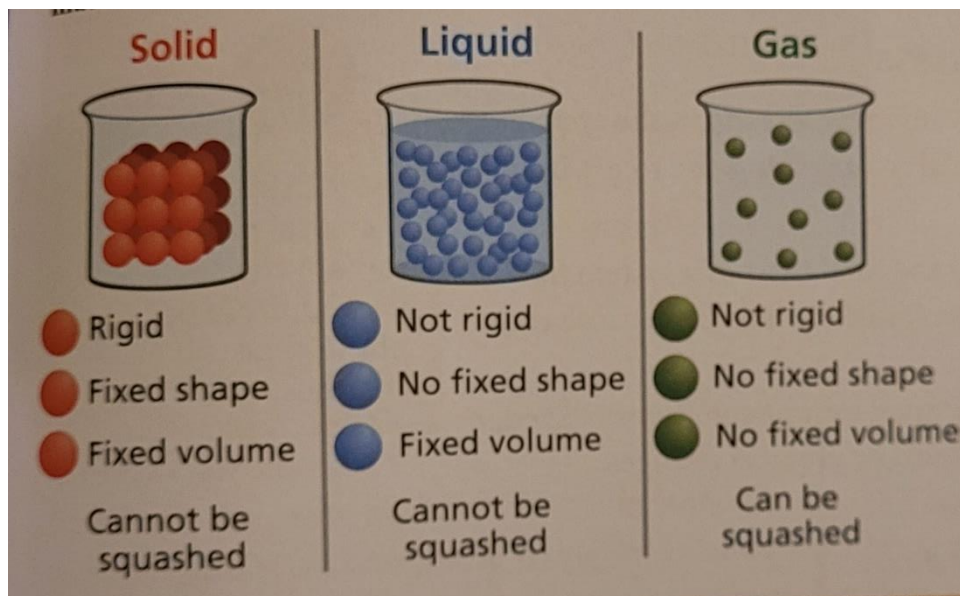
Weight=mass x force of gravity

Unit=Newtons

Atomic Nature of Matter

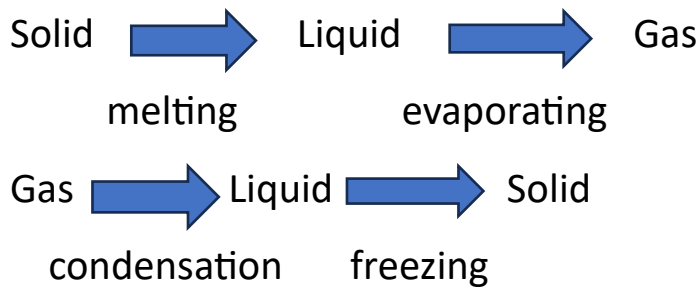
Atoms= are tiny particles that are building blocks of matter

Molecules= form when atoms join together.



Solid	Liquid	Gas
Particles are tightly packed together.	Particles stay close to one another but can move past each other.	Particles move in all directions and bounce off one another.
Particles can vibrate but stay close to one another.	Particles are able to flow.	Gas will spread out to fill any container it is held in.
Example: ice	Example: water	Example: water vapour

Changes of matter or state



Melting is changing of a **solid** into a **liquid** energy is gained.

Evaporating is a changing of a **liquid** into a **gas**.

Condensating is the changing of a **gas** into a **liquid**.

Freezing is the changing of a **liquid** into **solid** energy is removed.

Is there any difference between boiling and evaporating?

Evaporating occurs from **the surface of a liquid** it is **slow**. It happens **all the time**. It happens at **any temperature**.

Boiling occurs throughout **the liquid**. It is **fast** is only happens when the **liquid is heated** and given lots **of energy**. It happens at **100 degrees Celsius**.

What is meant by melting point temperature at which a solid turn into liquid.

Boiling point temperature at which liquid turns into a **gas** throughout the **liquid** **boiling point** of water is **100 Celsius**.

Diffusions is movement of particles from an area of high concentration to area of low concentration. Occurs mainly in liquids and gasses.

Elements compounds and mixtures

Elements

An element is a substance that cannot be split up into simpler substances by chemical means. Elements contain only one type of an atom.

All elements can be found in Periodic table

Elements	Symbol	Elements	Symbol
Hydrogen	H	Helium	He
Carbon	C	Neon	Ne
Nitrogen	N	Magnesium	Mg
Oxygen	O	Aluminium	Al
Phosphorus	P	Calcium	Ca

Compound

A compound is a substance that is made up of 2 or more. A different element that are chemically combined join together. Compounds contains two or more different types of atoms joined together by chemical bonds.

Water is an example of a compound because it contains 2 different elements, hydrogen and oxygen joined together by chemical bond. The formula for the water is H_2O .

Mixture

A mixture consists of two or more different substances, mainly with each other but not chemically combined

Elements in a mixture retain the individual properties of those elements. This is not the case with compounds.

Compound the elements in the compound are always present in the same amount.

Compound the elements in a compound are chemically combined.

Compound It's usually difficult to separate the elements in a compound. It is only possible to do using the suitable experiments.

Compound which is usually given out or taken in when a compound is formed.

Mixture and make sure contains two or more substances.

Mixture the substances in a mixture are not chemically combined.

Mixture is usually easy to separate the parts of a mixture.

Mixture there are almost no energy changes when a mixture is made.

Mixtures

Solutions

Soluble substances are able to dissolve in the liquid e.g. sugar in water.

Insoluble substances do not dissolve in liquid e.g. sand in water.

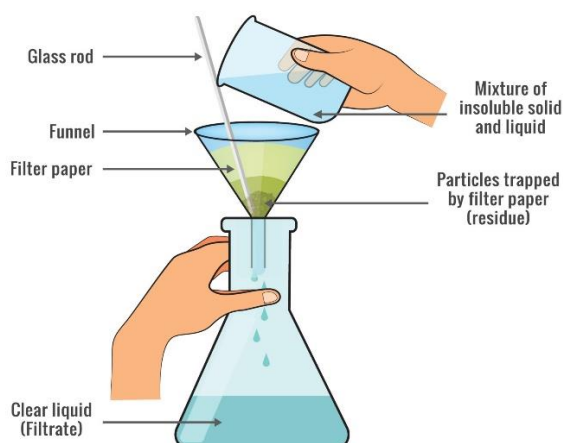
Solute a substance that dissolves in another substance e.g. sand

Solvent a substance that dissolves other substances e.g. water

1. Filtration

A good method of separation for solids that are insoluble in water.

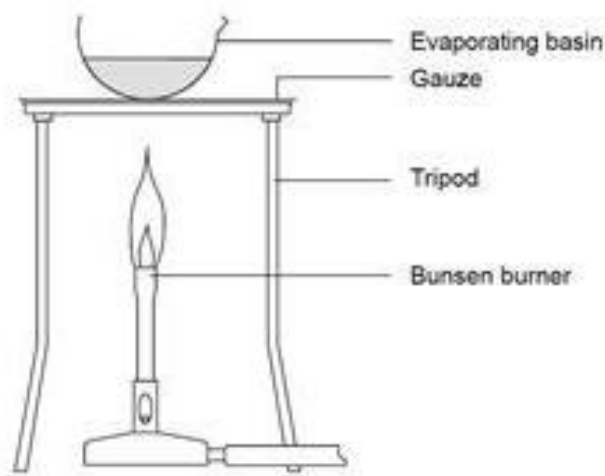
Filtration Process



Sand, water and salt separated by

2. Evaporation

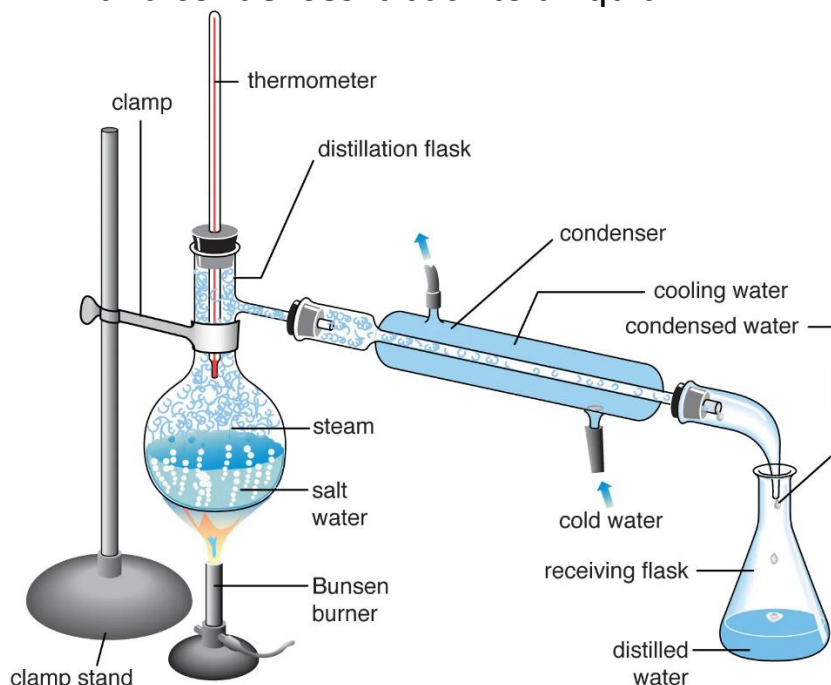
Appropriate for solids that are soluble in water like salt.



- Solution of water and salt
- Solution is heated to boiling point for the liquid to evaporate.
- Solid particles of salt remain after liquid evaporates.

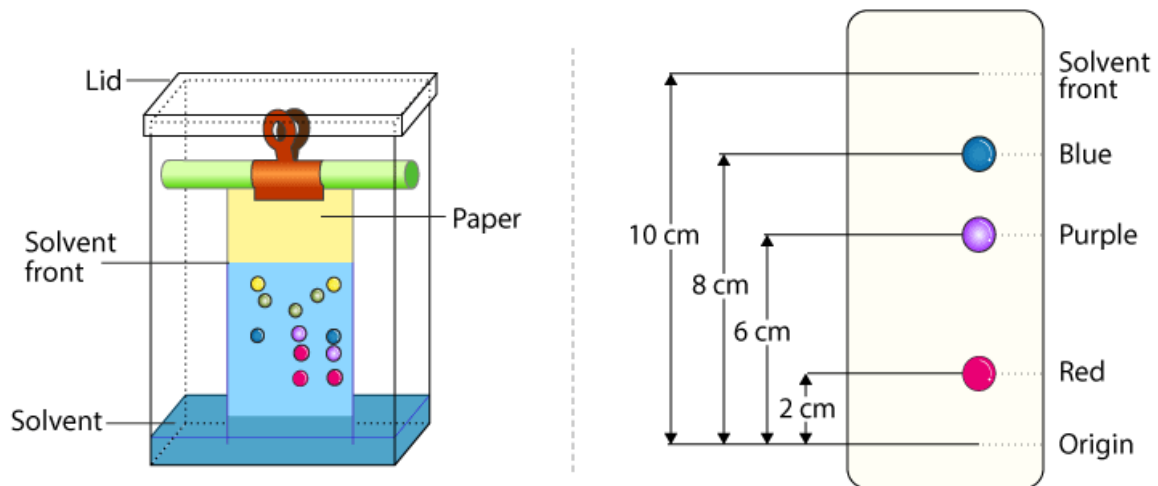
3. Distillation

- Useful when you want to keep the liquid part after separation.
- It can be used to separate two mixtures of liquids with different boiling points.
- A **Liebig condenser** traps gas and turns it back into a liquid.
- Inner tube allows gas to travel through.
- Outer tube lets cold water travel through which, cools the gas and condenses it back to a liquid.



4. Chromatography

- A way of separating mixtures made up of many different parts.
- Example: Black ink which is made of many different colours all together.
- Chromatography separates these inks because different ink particles travel different distance up the chromatography paper.



Space

Astronomy is the study of moon, stars, planets, comets, gas, galaxies and the universe.

An astronomer is person that studies astronomy.

Celestial bodies are naturally made objects outside the earth's atmosphere. They include stars planets, moons, asteroids and comets.

A satellite is a object orbiting a celestial body of larger mass.

A moon is the natural satellite of a planet.

A comet a dusty snowball that orbits a star.

Stars are celestial bodies.

They are luminous objects made up of gas that can make its own light and heat by nuclear.

Nuclear reactions

When two hydrogen atoms join together and fuse to make helium (Nuclear fusion).

A planet needs to be:

- Orbit the sun.
- Have its own gravity to make its sphere shape.
- Big enough that its gravity could move another object out of the way.

Terrestrial planets

- Mercury
- Venus
- Earth
- Mars

Juian Planets

- Jupiter
- Saturn
- Neptune
- Uranus

Galaxies

Galaxy a collection of billions of stars, gas and dust held together by gravity.

Billions of galaxies in the universe

Earth is in the milky way galaxy-this a spiral shape which is the most common shape for galaxies.

Edwin Hubble

- Hubble discovered the universe is expanding at a great rate.
- Hubble's Law: Galaxies are moving away from each other. The farther away a galaxy is faster the galaxy is moving from us.
- The Hubble Space Telescope was launched by NASA in 1990 and was named after Edwin Hubble
- Telescopes can be placed either on the ground or on satellites (orbiting or circling the Earth).

What's so special about the Hubble Space Telescope

When we look into space using a telescope on the earth the image collected tend to be quite fuzzy because the Earth's atmosphere dims, scatters and distorts the light.

The Hubble Space Telescope is in the space circulating the earth and is not affected by the Earth's atmosphere.

The Big Bang Theory

- Before the Big Bang over 13 billion years ago the entire Universe was compressed into a tiny hot mass
- A huge expansion allowed all of the Universe's matter and energy to form stars and galaxies.
- Gravity began to bring matter together to form stars and galaxies.
- At first the Universe expanded faster than the speed of light, before slowing down.
- Therefore, light generated around the time of the Big Bang is only now catching up with us here on Earth.

Energy

Energy is ability to do work.

(Joules)=measure

Kinetic= when an object is moving

Potential=stored in an object due to its change of shape or position you have given

Heat(thermal)= particles move (thermometer)

Chemical= stored in food, stored in batteries or coal

Solar= cause plants to grow

Sound=make vibrations.

Electrical= moving charges

Nuclear energy=stored in the nucleus of an atom. Atoms can be split (fission) or joined together.

Magnetic=conversion of energy to different forms.

Energy cannot be created or destroyed but it can from one form to another.

Biomass energy=chemical energy stored in plants(ratting of these plants produced methane gas,plants are burn to realise energy).

Source of energy

Sun is the primary source of energy photosynthesis, solar energy, heat.

Renewable energy= won't run out

Non renewable energy= will run out

Renewable

Solar

Hydro-electric

Wind

Water (waves and tidal)

Biomass

Geothermal

Non-renewable

Coal

Oil

Gas

Nuclear

Non-renewable sources of energy

Fossil fuels are made from the remains of dead plants and animals formed millions of years ago.

They are mainly composed of **two elements**.

1. Carbon.
2. Hydrogen.

Hydrocarbons are compounds made up of hydrogen and carbon.

Solar energy=energy from the sun converted into electricity.

Wind energy=wind turns the turbines the energy is harnessed and converted into electricity.

Hydroelectricity=water builds up behind a dam kinetic energy is generated when the water is released.

Wave energy=rocking motion of large floods by the used to generate electricity.

Geothermal energy=used to produce heat energy cold water pumped down one pipe and returns up the other as hot water or steam.

Tidal energy=high tide water builds up behind a dam on an estuary. At low tide water is let out again generating electricity.

Non-renewable energy sources

Advantages

- Relatively cheap and readily available
- Easy to use

Disadvantages

- Produce greenhouse gases (fossil fuels) burning gives off CO₂.
- Mining for fuels damages the environment.
- Getting are expensive rise in electricity costs.

Renewable Energy

Advantages

- Renewable source of energy will never be used up.
- Don't produce harmful waste products.

Disadvantages

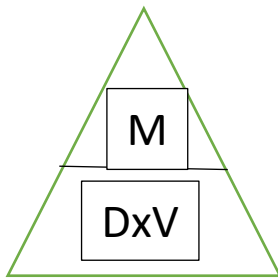
- Expensive to set up initially.
- Can be difficult to harvest the energy to create electricity.

Energy Conservation

- Wall insulation
- Attic insulation
- Double glazing windows
- Draft prevention
- Energy saving light bulbs.

Saves money on fuel bills as well as decreasing amount of fossil fuels burned better for the environment.

Density



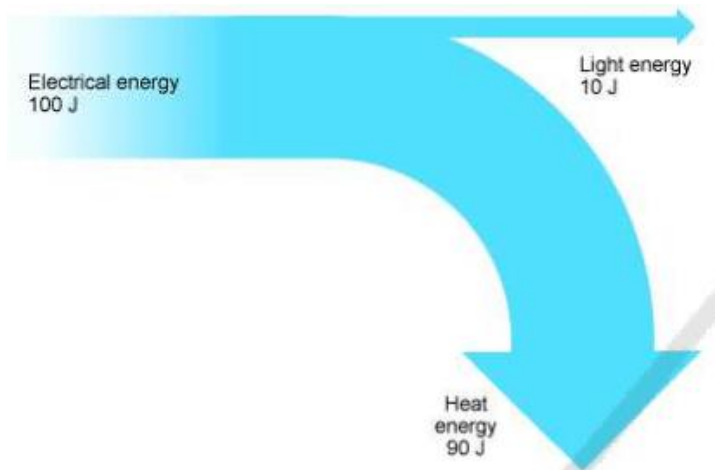
M is for mass

D is for density

V is for volume Mass (g) 58.3 g

Example: density = $\frac{\text{Mass (g)}}{\text{Volume (cm}^3\text{)}} = \frac{58.3 \text{ g}}{18 \text{ cm}^3} = 3.23 \text{ g/cm}^3$

Sankey Diagram



The line represents the total energy entering the appliance per second the type of energy and the amount of energy are labelled.

The straight though arrow represents the amount of energy used for the purpose intended e.g. making light in a bulb. The thickness of the arrow also represents the amount of energy.

The downward-facing arrow shows the amount of energy dissipated (wasted). The form of energy and the amount of energy are labelled the thickness of the arrow also represents the amount of energy. You may have more than one arrow. If more than one form of energy is wasted.

